

Tenure performance metrics — discussion memo

Created: 2026-09-01
Audience: Charles + COMPASS
Status: Living spec fork — not yet locked for Act III (p) or F-HERO
Context: Slow-start Q&A after tenure porch BDPs (overlap, distributions) and Q16/Q20 HERO. Clarifies what v0 **actually uses** vs what MBB last-ps does, and lists peer-pool / perf alternatives.
Related: `TENURE_hero_pipeline.md` · `TENURE_HERO_Campaign_Plan.md` · `tenure/tenure_pipeline/pool_metrics.py` · `tenure/documents/PANEL_CSV_GLOSSARY.md`

TL;DR

Question	v0 answer
Last-ps like MBB?	No — HERO collapses mean LOO over all assistant years ; BDP overlap uses all assistant person-years .
Cumulative tenure pubs?	No for pool / HERO / H_sort — we use <code>pubs_year</code> (annual). <code>pubs_cumulative</code> exists in panel but is unused in v0.
Who is in the peer pool?	<code>rank == assistant</code> only , OpenAlex-matched subset for quality; pool key = <code>(uni_slug, year)</code> (university × year in code; docs sometimes say “dept-year”).
All professors in dept?	Not in v0 — associates/full not in pool; true department field not in pool key today.

1. What perf metric are we using? (Not last-ps)

MBB reference (reigning hero)

- Grain:** one row per athlete = **last player-season** (“last-ps”).
- Perf:** PPM (or ladder metric) **z-scored within season**.
- Pool pressure:** `poolq_loo` on that cross-section.

Tenure v0 (locked HERO + porch BDPs)

Two grains — **neither is last-ps**:

Layer	Unit	X / ability
Pass A HERO (Q16 default, Q20 robustness)	One person	Mean <code>poolq_loo_mean</code> over assistant years with non-null LOO
Porch BDP / interval overlap / H_sort	Assistant person-year	<code>pubs_year</code> , z-scored within calendar year

`poolq_loo_mean` at year t (Cell 8):

$$\text{poolq_loo}(i, t) = \text{mean}(\text{pubs_year}(j, t))$$

for OA-matched assistants j in the same pool as i , $j \neq i$.
So HERO x-axis is **time-averaged annual peer pressure**, not “final cumulative pubs at exit.”

2. Annual pubs vs cumulative — and postdocs with prior work

Panel fields (OpenAlex)

Field	Meaning
<code>pubs_year</code>	Publications attributed to this person in calendar year
<code>pubs_cumulative</code>	Running total through <code>year</code> (person-level spell; see glossary for gap caveats)

v0 pool metrics, HERO, and overlap H_sort use `pubs_year` only.

Interpretation

- Annual** `pubs_year` $\approx$ “Who are my publishing peers **this year**?” — contemporaneous comparison on the assistant clock.
- Cumulative** $\approx$ “Who has the bigger **career stock**?” — confounded by **years as assistant** (`years_as_asst_so_far`) and time on market.

Arriving with publications already (postdoc CV)

The panel only observes people **once they appear as assistant** in scrape snapshots.

- Prior career productivity can show up as **high** `pubs_year` in **early assistant years** (continued output), not as a separate “CV at hire” covariate unless we build one.
- We do **not** currently strip pre-hire pubs or flag “first assistant year.”

3. Who is in the peer pool?

v0 (Cell 8 — pool_metrics.py)

- **Members:** rows with rank == assistant in that (uni_slug, year) .
- **Quality computed on:** OpenAlex-matched subset (openalex_id present).
- **Excluded from pool:** associate, full, adjunct, unknown rank (for pool membership; they may still appear elsewhere in panel).
- **Pool key in code:** (uni_slug, year) — university-wide assistant cohort that year, not department slug (department may be implied by scrape URL but is not the pool partition key today).

Docstrings say “dept-year”; implementation is uni × year. Worth keeping explicit in prose and in any Alex brief.

Charles’s question: all professors in the department annually?

Yes — that is a different peer environment, not a tweak of the current one.

Spec	Pool members	Perf	Question
A — v0 (locked HERO)	Assistants only	pubs_year (LOO mean → HERO)	Tenure-track cohort pressure among peers
B	Assistants only	pubs_cumulative	Sorting on stock among assistants (tenure clock confound)
C	All ranks (asst + assoc + full)	annual or cumulative	Full departmental publication environment
D	Assistants only	last assistant year only (“tenure last-ps”)	MBB-style exit cross-section

Spec C is often closer to “who do I compare myself to in the building?” but requires:

- roster definition for all ranks at uni/dept × year;
- perf rule for seniors (annual vs cumulative; active researcher filter);
- OpenAlex coverage asymmetry (seniors may be missing or mis-linked).

Not wired in v0. Treat A vs C as two environments (Alex: environment ≠ advancement), not one blended plot.

4. Link to homophily (H_sort) and p

Empirical sorting index (H_sort) = share of cross-person variance in z-scored perf explained by pool assignment (same formula as MBB Grandchild readout).

Porch overlap (infHM): H_sort ≈ 0.173 on pubs_year z within year, pools = uni × year, assistants + OA.

Homophily parameter (p) in Grandchild ASSIGN is not H_sort. p is calibrated so simulated ASSIGN reproduces empirical H_sort (PD21 recipe), using:

- ability vector (here: pubs_year z),
- empirical roster caps per pool,
- Grandchild stub assignment.

If we change perf (annual → cumulative) or pool (assistants only → all ranks), H_sort and p* both change — that is an Act III spec lock, not a cosmetic relabel.

Campaign policy: Do not assume tenure p ≈ 0 like MBB PPM; H_sort ≈ 0.17 already suggests nontrivial sorting on the v0 partition.

5. Recommended sequencing (Charles + COMPASS)

1. **Keep Q16 HERO + infHM BDPs on Spec A** until Alex sees shape — already locked.
2. **Q20** — inspect dip robustness only; same Spec A perf/pool.
3. **Before p bracket:** decide whether Act III repeats Spec A or adds Spec C (or B/D) as a parallel porch, not a replacement, unless Alex reprioritizes.
4. **F-HERO (Act II):** fixed Â slice + department mean pubs — orthogonal but should cite same pool/perf language as hero env.

6. Open decisions (YOU ARE HERE)

ID	Decision	Options	Default
P1	HERO / pool perf	<code>pubs_year</code> vs <code>pubs_cumulative</code> vs rate (pubs / years on clock)	<code>pubs_year</code> (v0)
P2	Pool membership	assistants only vs all ranks	assistants only (v0)
P3	Pool geography	uni × year vs true department × year	uni × year (code today)
P4	HERO grain	mean LOO over spell vs last assistant year	mean over spell (v0)
P5	p calibration partition	same as P1–P3 vs separate “environment” spec	defer until P1–P3 reviewed with Alex

Thread log

Date	Entry
2026-09-01	Initial memo from chat: last-ps vs tenure grains; annual vs cumulative; assistant-only vs full-dept pool; p link.